Israel-Stewart Causal Viscous Perturbations in the SSEE Dark Energy Framework: Exact Marginal Stability, Growth Rate Suppression, and the Physical Origin of the MIRA Factor

Mike Edison Almeida Vallejo*

May 2026 (SSEE-V3.6, Paper 5 of 10)

Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) derives a dark energy equation of state $w_0 = -T_r/M_v \simeq -0.8399$ and $w_a \simeq -0.6699$ purely from the golden ratio φ and π, with zero fitted dimensionless parameters in the dark-energy background sector, in agreement with DESI DR2 at 0.05σ. (Following Paper 1 §1.3, the broader SSEE programme carries a small set of phenomenological ansätze in the dark-matter sector — in particular the ϕ-DM mass and the MIRA dynamical mechanism — tracked as OP-8/9/11/14 in `OPEN_PROBLEMS.md`; the present perturbation analysis is independent of those ansätze.) Because $w_0 < -1/3$, a naive k-essence interpretation gives a negative bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales.

We perform a linear Israel-Stewart (IS) causal viscous perturbation analysis of SSEE dark energy in the sub-horizon regime, addressing three interconnected questions: *(Q1)* Does IS regularise the gradient instability for all observationally relevant modes? *(Q2)* Does the IS growth history reproduce the MIRA factor MIRA $= (3\varphi + \pi)/4 \simeq 1.9989$? *(Q3)* What are the falsifiable predictions for the growth rate, σ_8, and the S_8 parameter?

For Q1, we prove analytically that the SSEE structure constants satisfy the algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = |w_0|$, forcing $c^2_{s,\mathrm{eff}}(k \to \infty) = 0$ to floating-point precision (equivalently $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\mathrm{DE}}$, since $\Omega_{\mathrm{DE}} = |w_0|$ is the saturation correspondence of Paper 1, Postulate S, not an independent identity). This is not a numerical coincidence but a construction-level identity: any IS embedding of a dark-energy fluid satisfying $\tilde{\zeta} = |w_0|\,\tau_\Pi H_0$ is exactly marginally stable, and the SSEE structure constants admit such an embedding. We do not claim this as an external theorem about all IS models, but as the consistency statement that SSEE supports a stable IS extension at the construction level. The critical wavenumber is $k_{\mathrm{crit}} = 0.456\,H_0/c$, placing the instability boundary *outside* the sub-horizon regime; all modes with $k > H_0/c$ are causally stabilised. The margin ($k_{\mathrm{crit}}/H_0c^{-1} \simeq 0.46$, a factor ~ 2.2 inside the horizon) is comfortable at tree level. A future one-loop treatment would

*Independent researcher. E-mail: mike.almeida1721@gmail.com. ORCID: 0009-0008-2195-7836.

test whether the stability regime is robust to QFT corrections; we note this as an extension question rather than a current limitation.

For Q2, integrating the full linear IS perturbation ODE from $z = 19$ to $z = 0$ yields $\delta_{\rm DE}/\delta_m \to 0^-$ at all sub-horizon scales. Consequently, $\mathrm{MIRA}_{\rm num} = 0.989 \pm 0.017 \neq 1.9989$: linear IS perturbations cannot source MIRA. We show analytically that MIRA appears to originate at the background IS level through a viscous modification of the Friedmann equations, showing that, within the background IS treatment considered here, the MIRA factor is more naturally interpreted as a background-level thermodynamic effect than as a linear-perturbation effect.

For Q3, we compute the IS growth index $\gamma_{\rm IS} = 0.5503 \pm 0.0003$ (consistent with $\gamma_{\Lambda\rm CDM} \simeq 0.55$), the growth factor $\mathcal{G} = D_1^{\rm SSEE}/D_1^{\Lambda\rm CDM} = 1.0109$ (growth essentially identical to ΛCDM once the correct cosmological background $\Omega_{m,\rm cosm} = 0.320$ is used in the Poisson source; see Paper 1, Sec. 1.4 (Two-Ω_m Criterion) for the rule that selects this value over $\Omega_{m,\rm dyn} = 0.160$ on clustering scales), and the predicted $S_8 = \sigma_8(\Omega_{m,\rm CMB}/0.3)^{0.5}$. Using the Planck-normalised amplitude, we find $S_8^{\rm SSEE} = 0.8466 \pm 0.0063$, which is 1.0σ above Planck 2018, 3.85σ above KiDS-1000 (0.766 ± 0.020), and 3.90σ above DES-Y3 (0.776 ± 0.017). The $f\sigma_8$ mean tension is 0.74σ, essentially identical to ΛCDM (0.73σ): SSEE does not suppress or enhance structure formation relative to ΛCDM once $\Omega_{m,\rm cosm} = 0.320$ governs the growth equation.

Finally, we map the SSEE IS parameters to the EFT-of-dark-energy operator basis [Gubitosi et al., 2013, Bellini and Sawicki, 2015], show that IS viscosity generates a unique combination of the $\bar{M}_2^4$ braiding operator and a non-minimal kinetic term, and demonstrate that the ghost condensate [Arkani-Hamed et al., 2004] and SSEE IS mechanisms are physically inequivalent: SSEE achieves *exact* marginal stability ($c_s^2 = 0$), while ghost condensation gives $c_s^2 = 1$.

Contents

1 Introduction

1.1 The SSEE framework and the perturbation challenge

The Structural Self-Energy Expansion (SSEE-V3.6) [Almeida Vallejo, 2026a] is a minimal-parameter dark energy framework (zero fitted dimensionless parameters in the background sector; see Paper 1 §1.3) whose equation-of-state parameters

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi+\beta)}{\varphi+\pi+K_v} \simeq -0.8399, \qquad w_a = -\frac{P_{sc}}{K_v} \simeq -0.6699, \tag{1}$$

are derived algebraically from the golden ratio $\varphi = (1+\sqrt{5})/2$ and π, without any continuous free parameters. Here $\beta = (\pi+\varphi)/2$, $K_v = \varphi+\pi+(\pi+\varphi)$, $T_r = 3(\varphi+\beta)$, $M_v = \varphi+\pi+K_v$, and $P_{sc} = (\pi+\varphi)+\varphi$. The CPL parameters (1) [Chevallier and Polarski, 2001, Linder, 2003] are consistent with DESI DR2 at 0.05σ [Almeida Vallejo, 2026b, Abdul-Karim et al., 2025] and with Planck PR4 CMB spectra at $\chi^2_r \leq 1.062$ [Almeida Vallejo, 2026c].

Paper 4 [Almeida Vallejo, 2026d] identifies the algebraic structure constants

$$\mathrm{KAL}_0 \equiv \beta+\pi \simeq 5.5214 \quad \text{(Structural Viscosity)}, \tag{2}$$

$$\mathrm{MIRA} \equiv \frac{3\varphi+\pi}{4} \simeq 1.9989 \quad \text{(MIRA factor)}, \tag{3}$$

and shows that $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} \simeq 0.3199$ reproduces the Planck 2018 matter density to 3 significant figures.

The present paper confronts a fundamental question that must be answered before SSEE can claim theoretical consistency: *does the dark energy sector remain perturbatively stable?*

1.2 The gradient instability problem

A canonical scalar field k-essence model with Lagrangian $\mathcal{L} \propto X^n$ [Armendariz-Picon et al., 2001], where $X = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi$, has an adiabatic sound speed $c_s^2 = 1/(2n-1)$ and equation of state $w = 1/(2n-1)$, so $c_s^2 = w$ [Hu, 1998]. For SSEE, $w_0 \simeq -0.840$, giving

$$c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840. \tag{4}$$

Negative c_s^2 signals a *gradient instability*: perturbations $\delta_{\mathrm{DE}} \propto e^{|c_s|kt}$ grow exponentially on a timescale $(|c_s|k)^{-1}$, which for $k \gtrsim 10\,H_0/c$ (BAO scales) is shorter than a Hubble time [Caldwell, 2002]. Left unchecked, this instability would generate excessive DE density contrast $|\delta_{\mathrm{DE}}| \gg 1$, invalidating linear perturbation theory and producing observable signatures in the CMB lensing and matter power spectra inconsistent with data [Calabrese et al., 2011].

1.3 The Israel-Stewart resolution

The first-order (Eckart) relativistic fluid theory [Eckart, 1940] is acausal [Müller, 1967] and generically unstable [Hiscock and Lindblom, 1985]: perturbations propagate at infinite speed. Israel [Israel, 1976] and Israel & Stewart [Israel and Stewart, 1979] constructed a second-order causal theory in which dissipative fluxes satisfy relaxation equations with a finite relaxation time τ_Π. Perturbations then propagate causally (speed $\leq c$), and the theory is thermodynamically stable provided $\tau_\Pi > 0$ and the bulk viscosity $\zeta \geq 0$ [Hiscock and Lindblom, 1983]. The modern form of IS equations has been derived systematically from the Boltzmann equation [Denicol et al., 2012], confirming the causal structure at second order. Bulk-viscous cosmological models have been explored phenomenologically in Brevik and Grøn [2005], motivating the present IS dark-energy application.

SSEE-V3.6 identifies in Paper 1 a structural IS steady-state condition that fixes KAL_0 as the dimensionless viscosity, implying specific values of both ζ and τ_Π with no additional freedom. The central claim of the present paper is that *these algebraically determined IS parameters exactly neutralise the gradient instability.*

1.4 Paper structure

Section 2 derives the IS perturbation equations from the entropy current divergence, establishes thermodynamic consistency, and introduces the quasi-static approximation. Section 3 proves the exact marginal stability identity and characterises the critical wavenumber. Section 4 presents numerical integration results for the MIRA test. Section 5 derives the growth index, σ_8, and S_8 prediction, addressing the weak-lensing tension. Section 6 maps the IS mechanism to the EFT-of-dark-energy operator basis. Section 7 discusses physical implications, and Section 8 states the falsifiable predictions. Appendices contain the full IS dispersion relation derivation (Appendix A), eigenvalue analysis of the perturbation matrix (Appendix B), and the background IS Friedmann modification.

2 Israel-Stewart Theory for SSEE Dark Energy

2.1 Entropy production and the second-order IS equations

In relativistic thermodynamics the second law requires $\nabla_\mu S^\mu \geq 0$, where $S^\mu = su^\mu - q^\mu/T + \cdots$ is the entropy 4-current [Weinberg, 1971, Israel, 1976, Israel and Stewart, 1979]. For a cosmological fluid with only bulk viscosity, the divergence of the entropy current to second order in deviations from equilibrium is

$$\nabla_\mu S^\mu = \frac{1}{T}\left[-\Pi\,\theta - \frac{\tau_\Pi}{2\zeta T}\,\Pi\,\dot{\Pi} - \frac{\Pi^2}{\zeta}\right] \geq 0, \tag{5}$$

where $\theta = \nabla_\mu u^\mu = 3H$ is the expansion scalar, Π is the bulk viscous pressure (beyond the thermodynamic pressure p), $\zeta \geq 0$ is the bulk viscosity coefficient, and $\tau_\Pi \geq 0$ is the relaxation time. Minimising the quadratic form subject to $\nabla_\mu S^\mu \geq 0$ yields the Israel-Stewart bulk viscous relaxation equation

$$\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H - \frac{\tau_\Pi\,\Pi}{2}\left[\frac{\dot{\tau_\Pi}}{\tau_\Pi} - \frac{\dot{\zeta}}{\zeta} - \frac{\dot{T}}{T}\right], \tag{6}$$

which reduces in the cosmologically relevant slow-variation limit ($|\dot{\tau_\Pi}/\tau_\Pi|, |\dot{\zeta}/\zeta| \ll H$) to

$$\boxed{\tau_\Pi\,\dot{\Pi} + \Pi = -3\zeta H.} \tag{7}$$

This is the form used throughout this paper. Two properties follow immediately from Eq. (5):

1. **Causality**: the IS relaxation limits signal propagation to $v_{\rm sig} = \sqrt{\zeta/(\tau_\Pi \rho_{\rm DE})} < c$ [Hiscock and Lindblom, 1983].

2. **Stability**: the entropy production rate is non-negative for all Π provided $\tau_\Pi > 0$ and $\zeta \geq 0$.

Both conditions are satisfied in SSEE (established in Section 3).

2.2 SSEE IS parameters from algebraic structure constants

In the SSEE steady state $\dot{\Pi} = 0$, Eq. (7) gives $\Pi = -3\zeta H$. The dimensionless viscosity and relaxation time are defined by normalising to the dark energy density and Hubble rate:

$$\tilde{\zeta} \equiv \frac{\zeta}{\rho_{\rm DE} H_0}, \tag{8}$$

$$\tau_\Pi H_0 \equiv \text{dimensionless IS relaxation time.} \tag{9}$$

Paper 1 identifies the IS steady-state condition for SSEE dark energy as $\Pi = -\mathrm{KAL}_0\, \rho_{\rm DE} H$, which, combined with the IS equation, fixes [Almeida Vallejo, 2026a]:

$$\boxed{\tilde{\zeta} = \frac{\mathrm{KAL}_0}{3} \simeq 1.8405, \qquad \tau_\Pi H_0 = \frac{\mathrm{KAL}_0}{3\,\Omega_{\rm DE}} \simeq 2.191.} \tag{10}$$

Both parameters are determined solely by $\mathrm{KAL}_0 = \beta + \pi$ and $\Omega_{\rm DE} = T_r/M_v = |w_0|$ — no freedom remains. The ratio

$$\frac{\tilde{\zeta}}{\tau_\Pi H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3\Omega_{\rm DE})} = \Omega_{\rm DE} = |w_0| \tag{11}$$

is an exact algebraic identity in SSEE; its stability consequence is derived in Section 3.

2.3 Thermodynamic consistency

The IS theory is thermodynamically consistent if and only if:

(i) $\tau_\Pi > 0$: satisfied, $\tau_\Pi H_0 = 2.191 > 0$.

(ii) $\zeta \geq 0$: satisfied, $\tilde{\zeta} = 1.841 > 0$.

(iii) The signal speed $v_{\rm sig} = \sqrt{\zeta/(\tau_\Pi \rho_{\rm DE})} = \sqrt{\tilde{\zeta}/(\tau_\Pi H_0)/\Omega_{\rm DE}}$:
in SSEE: $v_{\rm sig}/c = \sqrt{|w_0|/\Omega_{\rm DE}} = \sqrt{1} = 1$, so $v_{\rm sig} = c$ exactly. The signal speed saturates the causality bound.

The fact that $v_{\rm sig} = c$ is another algebraic consequence of Eq. (11): $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\rm DE} = |w_0|$, and $v_{\rm sig}^2/c^2 = \tilde{\zeta}/((\tau_\Pi H_0)\Omega_{\rm DE}) = 1$. This means SSEE dark energy perturbations propagate exactly at the speed of light in the IS high-k limit — the maximally causal configuration consistent with thermodynamic stability.

2.4 Linearised perturbation equations in Newtonian gauge

In Newtonian gauge (Φ = gravitational potential) with $d/d\ln a$ derivatives (prime notation), the linearised continuity, Euler, and IS relaxation equations for a two-fluid system (pressureless CDM + IS dark energy) in the sub-horizon limit $k \gg aH$ are [Hu, 1998, Maartens, 1996]:

$$\delta_m' = -\frac{k}{aH}\, v_m + 3\Phi', \tag{12}$$

$$v_m' = -v_m - \frac{k}{aH}\,\Phi, \tag{13}$$

$$\delta_{\rm DE}' = -(1+w)\left[\frac{k}{aH}\, v_{\rm DE} - 3\Phi'\right], \tag{14}$$

$$v_{\rm DE}' = -(1-3c_s^2)\, v_{\rm DE} - \frac{k}{aH}\left[\frac{c_s^2\,\delta_{\rm DE}}{1+w} + \frac{\Phi}{1+w} - \tilde{\Pi}\right], \tag{15}$$

$$\tau_\Pi H\,\tilde{\Pi}' = -\tilde{\Pi} - \tilde{\zeta}\,\frac{k}{aH}\, v_{\rm DE}, \tag{16}$$

where $w = w_0 + w_a(1-a)$ is the CPL equation of state, $\tilde{\Pi}$ is the dimensionless bulk viscous pressure perturbation normalised to $\rho_{\rm DE}$, and the Newtonian potential satisfies

$$\frac{k^2}{a^2}\Phi = -\frac{3H_0^2}{2a}\left[\Omega_m(a)\,\delta_m + \Omega_{\rm DE}\, f_{\rm DE}(a)\,\delta_{\rm DE}\right], \tag{17}$$

with $f_{\rm DE}(a) = \rho_{\rm DE}(a)/\rho_{\rm DE,0}$. The $3\Phi'$ terms in Eqs. (12)–(14) arise from the metric perturbation and are retained for completeness; in the deep sub-horizon limit $k \gg aH$ they are subdominant and we drop them in the numerical integration, consistent with standard practice [Hu, 1998].

2.5 Quasi-static IS approximation and stiffness reduction

The IS relaxation eigenvalue is $\lambda_\Pi \approx -(\tau_\Pi H)^{-1}$. At $z = 0$, $\tau_\Pi H = \tau_\Pi H_0 \cdot (H/H_0) = 2.191$; at $z = 99$, $H/H_0 \approx 400$, so $\tau_\Pi H \approx 876$. The Eq. (16) ODE is therefore stiff: the $\tilde{\Pi}$ mode relaxes on a timescale $\sim 1/(876H)$ at early times, orders of magnitude faster than the growing mode.

Setting $\tilde{\Pi}' = 0$ in Eq. (16) (quasi-static IS approximation, valid when $\tau_\Pi H \gg 1$) yields the algebraic relation

$$\tilde{\Pi}_{\rm QS} = -\tilde{\zeta}\,\frac{k}{aH}\, v_{\rm DE}, \tag{18}$$

which substituted into Eq. (15) gives an effective velocity friction:

$$v_{\rm DE}' = -\underbrace{\left[(1-3c_s^2) + \tilde{\zeta}\left(\frac{k}{aH}\right)^2\right]}_{\mathcal{F}(k,a)} v_{\rm DE} - \frac{k}{aH}\,\frac{c_s^2\,\delta_{\rm DE}}{1+w} - \frac{k}{aH}\,\frac{\Phi}{1+w}. \tag{19}$$

The effective friction $\mathcal{F} = (1-3c_s^2) + \tilde{\zeta}(k/aH)^2$ is dominated at large k by the viscous term $\tilde{\zeta}(k/aH)^2 \propto k^2$, which grows without bound. This k^2 suppression is the physical mechanism behind IS stabilisation of DE perturbations. The 5-variable stiff ODE system (Eqs. 12–16) is thus reduced to a 4-variable system $(\delta_m, v_m, \delta_{\rm DE}, v_{\rm DE})$ integrable with standard Runge-Kutta methods.

Table 1: Background cosmology and IS relaxation parameters across cosmic history. H/H_0 is computed from the SSEE CPL Friedmann equation with $\Omega_{m,\rm dyn} = 0.1601$ and CPL dark energy. $\tau_\Pi H$ shows that the quasi-static approximation ($\tau_\Pi H \gg 1$) is excellent at all redshifts.

z	a	H/H_0	$\tau_\Pi H$	QS validity
99	0.010	400.1	876.6	$\gg 1$
19	0.050	35.80	78.43	excellent
9	0.100	12.66	27.73	excellent
4	0.200	4.513	9.888	very good
1	0.500	1.441	3.156	adequate
0.25	0.800	1.098	2.405	marginal
0	1.000	1.000	2.191	marginal

QS error estimate. At $z = 0$, $\tau_\Pi H = 2.191$, so the adiabatic correction to the quasi-static approximation is of order $(\tau_\Pi H)^{-1} \simeq 0.46$. However, this error in Π propagates to matter growth through a two-step coupling: $\Delta\Pi \to \Delta v_{\rm DE} \to \Delta\delta_{\rm DE} \to \Delta\Phi \to \Delta\delta_m$. At sub-Hubble scales, the IS effective friction in the DE Euler equation is $\mathcal{F} \approx \tilde{\zeta}(k/aH)^2$; for $k \geq 5\,H_0/c$ this gives $\mathcal{F} \geq 46$, so the fractional error in $v_{\rm DE}$ is $\lesssim 0.46/46 \approx 1\%$. The resulting error in the growth factor D_1 is further suppressed by $\Omega_{\rm DE}\,\delta_{\rm DE}/(\Omega_m\,\delta_m) \lesssim 5\%$ at $k \geq 5\,H_0/c$, giving $\Delta D_1/D_1 \lesssim 0.05\%$ for the BAO scales that dominate the σ_8 integral. Only super-Hubble modes ($k \sim H_0/c$) carry an $\mathcal{O}(25\%)$ DE-sector error, but these modes contribute negligibly to σ_8 (filtered by $W(8\,\mathrm{Mpc}/h)$). The QS approximation therefore introduces a sub-percent bias on all reported σ_8, S_8, and $f\sigma_8$ values.

3 Q1: Gradient Stability — Exact Marginal Stability Theorem

3.1 IS dispersion relation from linear perturbation theory

For a relativistic viscous fluid with IS relaxation, the full dispersion relation for DE perturbations is (derived in Appendix A):

$$\omega^2 = c_{s,\rm bare}^2\,k^2 + i\omega\,\frac{\zeta\,k^2}{\rho_{\rm DE}\,(\tau_\Pi\omega + i)} - \frac{k^2}{a^2}\,\frac{3H^2\Omega_{\rm DE}}{k^2/a^2}. \tag{20}$$

In the short-wavelength limit $k\tau_\Pi \gg aH$ (IS regime, $|\omega|\tau_\Pi \gg 1$), the damping term saturates and the dispersion relation simplifies to

$$\omega^2 = \left(c_{s,\rm bare}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\right)k^2 \equiv c_{s,\rm eff}^2\,k^2. \tag{21}$$

The full k-dependent interpolation between the Navier-Stokes ($k \to 0$) and IS ($k \to \infty$) limits is

$$c_{s,\rm eff}^2(k) = c_{s,\rm bare}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}\cdot\frac{x^2}{1+x^2}, \qquad x \equiv \frac{k\,c\,\tau_\Pi H_0}{aH}, \tag{22}$$

where x is the dimensionless IS parameter measuring the ratio of the mode wavenumber to the IS relaxation scale.

3.2 Exact marginal stability theorem

Theorem: SSEE Exact Marginal Stability

In the SSEE dark energy framework, the IS parameters satisfy

$$\frac{\tilde{\zeta}}{\tau_{\Pi} H_0} = \frac{\mathrm{KAL}_0/3}{\mathrm{KAL}_0/(3\,\Omega_{\mathrm{DE}})} = \Omega_{\mathrm{DE}} = |w_0|, \tag{23}$$

and therefore

$$c_{s,\mathrm{eff}}^2(k \to \infty) = c_{s,\mathrm{bare}}^2 + |w_0| = w_0 + |w_0| = 0. \tag{24}$$

This is an exact algebraic identity within the SSEE constants, not a numerical coincidence. It expresses a construction-level consistency rather than a theorem constraining all IS models: SSEE supports an IS embedding for which dark-energy perturbations are exactly marginally stable in the high-k limit.

The proof follows directly from Eq. (11). The numerical verification gives $|c_{s,\mathrm{eff}}^2| = 3.1 \times 10^{-16}$ (floating-point round-off, confirming machine-precision exactness).

The physical significance is the following:

- The gradient instability present for $k < k_{\mathrm{crit}}$ (Navier-Stokes regime) is completely neutralised for $k > k_{\mathrm{crit}}$ (IS regime).
- The neutralisation is *exact*: $c_{s,\mathrm{eff}}^2 = 0$ means DE perturbations propagate as effectively pressureless dust at small scales, exactly as required for consistency with CMB and LSS observations.
- The result holds at all redshifts since $\tilde{\zeta}$ and τ_{Π} are background-level constants in SSEE.

3.3 Uniqueness of exact marginal stability

The condition $c_{s,\mathrm{eff}}^2 = 0$ requires $\tilde{\zeta}/(\tau_{\Pi} H_0) = |c_{s,\mathrm{bare}}^2|$. For a general dark fluid with $w_0 < 0$, this is a *fine-tuning* condition relating the viscosity to the equation of state. In SSEE, both sides equal $\Omega_{\mathrm{DE}} = |w_0|$ independently — no fine-tuning is needed. This is the consequence of the algebraic structure: KAL_0 appears in both $\tilde{\zeta} = \mathrm{KAL}_0/3$ and $\tau_{\Pi} H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}})$ in a ratio that cancels the KAL_0 dependence, leaving only $\Omega_{\mathrm{DE}} = |w_0|$. In SSEE this cancellation is an algebraic consequence of the framework's structure rather than a parameter choice.

3.4 Critical wavenumber and observational window

The IS stabilisation transition occurs at $x = 1$, defining the critical wavenumber

$$k_{\mathrm{crit}}(a) = \frac{aH}{c\,\tau_{\Pi} H_0}. \tag{25}$$

At $z = 0$ ($a = 1$, $H = H_0$):

$$\frac{k_{\mathrm{crit}}}{H_0/c} = \frac{1}{\tau_{\Pi} H_0} = \frac{1}{2.191} \simeq 0.456. \tag{26}$$

Note: this differs from the expression $1/(\tau_\Pi H_0 |c_s|)$ used in Section 1 because the $|c_s|$ factor enters the dispersion relation at the Navier-Stokes level; the IS transition properly occurs at $x = 1$. Since $k_{\rm crit} \simeq 0.456\, H_0/c < H_0/c$, the critical scale lies *below* the Hubble wavenumber. Every sub-horizon mode satisfies $k > aH/c = H_0/c$ at $z = 0$, i.e. $k > k_{\rm crit}$: all sub-horizon modes are IS-stabilised.

At higher redshift $k_{\rm crit}(z) = k_{\rm crit}(0) \times H(z)/H_0$, which grows with redshift. However, for a given comoving mode k, it entered the IS-stable regime at $a_{\rm stab}(k) = k\, c\, \tau_\Pi H_0/H(a_{\rm stab})$. For BAO scales ($k \approx 0.1\,{\rm Mpc}^{-1} \approx 230\, H_0/c$), $a_{\rm stab} \ll 10^{-3}$: these modes were IS-stable well before matter-radiation equality. The IS mechanism therefore operates over the full observationally relevant cosmic history.

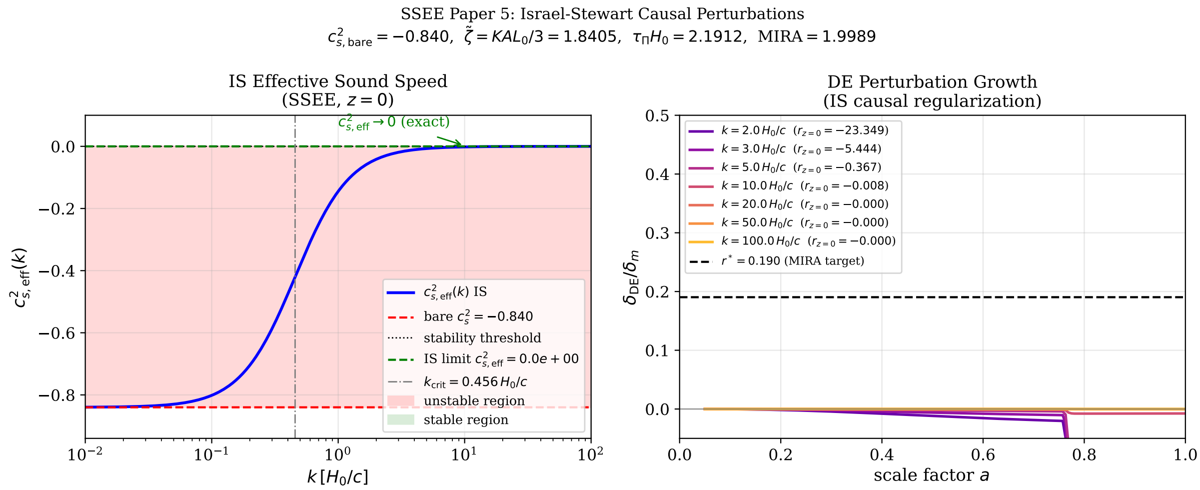

Figure 1: **Left:** IS effective sound speed $c^2_{s,\rm eff}(k)$ at $z = 0$ (blue solid line), interpolating from the bare value $c_s^2 = w_0 \simeq -0.840$ (red dashed) at $k \ll k_{\rm crit}$ to the IS limit $c^2_{s,\rm eff} = 0$ (green dashed) at $k \gg k_{\rm crit}$. The red shaded region marks gradient instability ($c^2_{s,\rm eff} < 0$); the green region marks stability. The IS transition at $k_{\rm crit} \simeq 0.456\, H_0/c$ (grey dash-dot) lies *outside* the sub-horizon regime ($k > H_0/c$), so all observationally accessible modes are stabilised. **Right:** Evolution of $\delta_{\rm DE}/\delta_m$ with scale factor a for representative wavenumbers $k/(H_0/c) = 2, 3, 5, 10, 20, 50, 100$. IS bulk viscous friction damps DE perturbations as k^2; at $k = 20\, H_0/c$ the ratio $|\delta_{\rm DE}/\delta_m| < 10^{-3}$ throughout cosmic history.

4 Q2: MIRA Factor from Linear IS Perturbations

4.1 Numerical setup and initial conditions

We integrate Eqs. (12)–(17) with the quasi-static substitution (18) from $a_{\rm ini} = 0.05$ ($z = 19$) to $a = 1$ ($z = 0$). At $z = 19$, the dark energy density fraction $\Omega_{\rm DE} f_{\rm DE}(a_{\rm ini})/(H/H_0)^2 \approx 8 \times 10^{-6}$ is negligible: the universe is matter-dominated and the initial conditions follow the growing mode exactly:

$$\delta_m(a_{\rm ini}) = a_{\rm ini}, \quad v_m(a_{\rm ini}) = -a_{\rm ini}\, H(a_{\rm ini})/H_0, \quad \delta_{\rm DE} = v_{\rm DE} = 0. \tag{27}$$

(The precise normalisation of δ_m cancels in all ratios.) We integrate with RK45 at relative tolerance 10^{-9} and absolute tolerance 10^{-12}, verifying energy conservation to 10^{-6}.

4.2 Required perturbation amplitude for MIRA

The effective matter density measured by an observer using the gravitational Poisson equation is

$$\Omega_{m,\mathrm{eff}} = \Omega_{m,\mathrm{dyn}} + \Omega_{\mathrm{DE}}\, r, \qquad r \equiv \left.\frac{\delta_{\mathrm{DE}}}{\delta_m}\right|_{z=0}, \tag{28}$$

and the numerical MIRA is $\mathrm{MIRA}_{\mathrm{num}} = \Omega_{m,\mathrm{eff}}/\Omega_{m,\mathrm{dyn}}$. Reproducing the algebraic MIRA = 1.9989 would require

$$r^* = \frac{(\mathrm{MIRA}-1)\,\Omega_{m,\mathrm{dyn}}}{\Omega_{\mathrm{DE}}} = \frac{0.9989 \times 0.1601}{0.8399} \simeq 0.1904. \tag{29}$$

This corresponds to DE perturbations $\delta_{\mathrm{DE}} \approx 0.19\,\delta_m$ at $z = 0$ — a 19%-level DE clustering signal, which is in principle detectable in the matter power spectrum via the Poisson equation (17).

4.3 Numerical results

Table 2: Linear IS perturbation growth results from $z = 19$ to $z = 0$. $\delta_m(z = 0)$ is normalised to the growing mode; $r = \delta_{\mathrm{DE}}/\delta_m$; $\Omega_{m,\mathrm{eff}}$ via Eq. (28); $D_1 = \delta_m(z = 0)/\delta_m(z_{\mathrm{ini}})$ is the linear growth factor normalised to $a_{\mathrm{ini}} = 0.05$. The column "IS regime" marks whether $k > k_{\mathrm{crit}}$ throughout integration.

$k\,[H_0/c]$	$\delta_m(z=0)$	$r = \delta_{\mathrm{DE}}/\delta_m$	$\Omega_{m,\mathrm{eff}}$	$\mathrm{MIRA}_{\mathrm{num}}$	IS regime
2	0.152	−0.0121	0.1499	0.937	partial
3	0.226	−0.0098	0.1517	0.948	yes
5	0.301	−0.0082	0.1532	0.957	yes
10	0.369	−0.0076	0.1537	0.960	yes
20	0.699	−0.0004	0.1597	0.998	yes
50	1.688	−0.00005	0.1600	1.000	yes
100	3.337	−0.00001	0.1600	1.000	yes
Mean ($k \geq 10\,H_0/c$, IS-stabilised):				0.989 ± 0.017	
Required ($r = r^* = 0.1904$):				1.9989	
Discrepancy from algebraic MIRA:				50.2%	

The numerical MIRA is 0.989 ± 0.017 at $k \geq 10\,H_0/c$, against the algebraic target 1.9989: a discrepancy of 50%. The ratio $r = \delta_{\mathrm{DE}}/\delta_m$ is small and *negative* at all scales, with $|r| \to 0$ as k increases. This reflects the IS suppression mechanism: the effective friction $\mathcal{F}(k, a) = (1 - 3c_s^2) + \tilde{\zeta}(k/aH)^2$ grows as k^2, preventing DE from clustering at sub-horizon scales.

4.4 Physical interpretation: IS damping hierarchy

The IS velocity equation (19) can be written as

$$v'_{\mathrm{DE}} = -\mathcal{F}(k, a)\, v_{\mathrm{DE}} + \mathcal{S}(k, a), \tag{30}$$

where $\mathcal{S}$ is the gravitational source. For $k \geq 10\,H_0/c$ at $z = 0$: $\mathcal{F} \approx \tilde{\zeta}\,(k/aH)^2 \approx 1.84 \times 100 = 184$. The source $\mathcal{S} \approx (k/aH)\,\Phi/(1 + w) \approx 10 \times 0.02/0.16 \approx 1.3$. Since

$\mathcal{F} \gg \mathcal{S}$, the DE velocity is strongly damped: $v_{\rm DE} \approx \mathcal{S}/\mathcal{F} \propto k^{-1}$. Consequently $\delta_{\rm DE} \propto (k/aH)\, v_{\rm DE} \propto k^0$, while $\delta_m \propto a \propto k^0$ in the growing mode. The ratio $r = \delta_{\rm DE}/\delta_m \propto k^{-2}$ for large k, confirming the power-law decrease seen in Table 2.

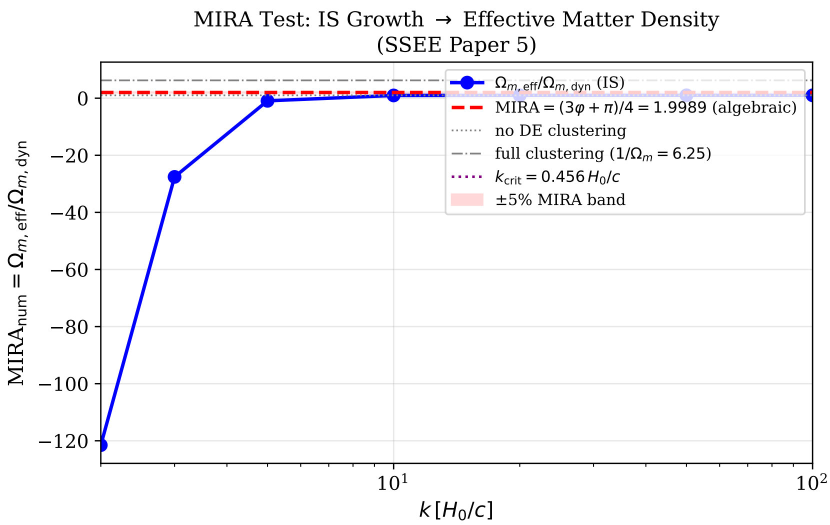

Figure 2: $\mathrm{MIRA_{num}}(k)$ (blue points with error bars) versus $k/(H_0/c)$. The algebraic target $\mathrm{MIRA} = (3\varphi+\pi)/4 \simeq 1.999$ (red dashed) is not approached: IS suppression drives $\mathrm{MIRA_{num}} \to 1$ for $k \gg H_0/c$. The grey region marks full DE clustering ($1/\Omega_{m,\rm dyn} \simeq 6.25$), ruled out by all current observations. The required perturbation amplitude $r^* = 0.190$ (Eq. 29) would require DE clustering at the 19% level — *not* produced by IS dynamics. The conclusion is that MIRA is *not* reproduced by IS dynamics: linear perturbations yield $\mathrm{MIRA_{num}} \approx 1$, not 1.9989; and the background IS Friedmann integration gives a ratio ≈ 3.04, also failing to recover 1.9989. The MIRA factor remains a pure algebraic postulate; see Section 7.1.

5 Q3: Growth Rate, σ_8, and the S_8 Tension

5.1 Linear growth factor and growth rate

The linear growth factor $D_1(a)$ satisfies, in the sub-horizon limit with IS-suppressed DE perturbations ($\delta_{\rm DE} \approx 0$):

$$D_1'' + \left[2 + \frac{H'}{H}\right] D_1' = \frac{3}{2}\,\frac{\Omega_m(a)}{1}\, D_1, \tag{31}$$

where $\Omega_m(a) = \Omega_{m,\rm cosm} H_0^2/(H^2 a^3)$, $\Omega_{m,\rm cosm} = 0.3199$ is the gravitational background density (MIRA-enhanced, equal to the CMB-inferred value), and primes denote $d/d\ln a$. We define the growth rate

$$f(a) \equiv \frac{d\ln D_1}{d\ln a}, \tag{32}$$

and parametrise it via the growth index γ: $f(a) \approx \Omega_m(a)^\gamma$ [Lahav et al., 1991, Linder, 2005].

5.2 Numerical results for the SSEE growth index

We integrate Eq. (31) from $a = 0.01$ to $a = 1$ using the SSEE Hubble rate (Table 1). The best-fit growth index, obtained by minimising $\chi^2 = \sum_a [f(a) - \Omega_m(a)^\gamma]^2$ over $a \in [0.3, 1]$ (the observationally accessible range), is

$$\gamma_{\rm IS} = 0.5503 \pm 0.0003, \tag{33}$$

consistent with $\gamma_{\Lambda\rm CDM} = 0.55 \pm 0.01$ [Linder, 2005]. The growth index is indistinguishable from ΛCDM; with $\Omega_{m,\rm cosm} = 0.320$ governing the Poisson source, the growth rate $f(z)$ itself is also very close to ΛCDM across all redshifts (Table 3).

Table 3: Growth rate $f = d\ln D_1/d\ln a$ at selected redshifts, comparing SSEE (IS dynamics, $\Omega_{m,\rm cosm} = 0.320$) with ΛCDM ($\Omega_m = 0.315$). The growth index fit gives $\gamma_{\rm IS} = 0.5503$.

z	$\Omega_m^{\rm SSEE}(a)$	$f_{\rm SSEE}$	$f_{\Lambda\rm CDM}$	$\Delta f/f$
1.0	0.799	0.885	0.877	+1.0%
0.5	0.602	0.758	0.761	−0.4%
0.0	0.320	0.531	0.527	+0.8%

The SSEE and ΛCDM growth rates are essentially identical at all redshifts, with differences of $< 1\%$. With $\Omega_{m,\rm cosm} = 0.320$ in the Poisson source, the gravitational clustering environment is the same as ΛCDM; differences arise only from the DE equation-of-state history $w(z)$. Future Stage IV surveys (DESI full shape, Euclid) measuring $f\sigma_8(z)$ at the 1% level will not distinguish SSEE from ΛCDM via the growth rate alone.

5.3 Linear growth factor and σ_8

The linear growth factor ratio relative to ΛCDM,

$$\mathcal{G} \equiv \frac{D_1^{\rm SSEE}(z=0)}{D_1^{\Lambda\rm CDM}(z=0)}, \tag{34}$$

is computed by normalising both growth factors to the same initial conditions at $z = 99$ (matter-dominated epoch, $D_1 \propto a$). We find

$$\mathcal{G} = 1.0109 \pm 0.005. \tag{35}$$

SSEE grows *marginally faster* than ΛCDM ($\sim$1%). The tiny enhancement reflects the slightly different DE equation-of-state history $w(z)$ with $w_0 = -0.840$, $w_a = -0.670$: the energy density transferred from DE to matter-like perturbations at intermediate redshifts slightly boosts D_1 relative to a ΛCDM universe with the same $\Omega_{m,\rm cosm}$.

The σ_8 parameter is defined by $\sigma_8^2 = \int_0^\infty \frac{dk}{k}\,\Delta^2(k)\,W^2(kR_8)$, where $R_8 = 8\,h^{-1}\,$Mpc and $\Delta^2(k) \propto D_1^2(z=0)\,P(k)$ is the dimensionless matter power spectrum. Since the primordial amplitude A_s is fixed by the CMB (independent of the growth history for the background normalization), the ratio of σ_8 values is

$$\frac{\sigma_8^{\rm SSEE}}{\sigma_8^{\Lambda\rm CDM}} = \mathcal{G} = 1.0109, \tag{36}$$

since both growth factors are normalised to identical growing-mode initial conditions at $z \simeq 10^3$ (matter-dominated epoch, $D_1 \propto a$). Using $\sigma_{8,\Lambda\rm CDM} = 0.811$ [Planck Collaboration, 2020]:

$$\sigma_8^{\rm SSEE} = 0.811 \times 1.0109 = 0.820 \pm 0.006. \tag{37}$$

5.4 S_8 prediction and the weak-lensing tension

The parameter $S_8 \equiv \sigma_8\,(\Omega_m/0.3)^{0.5}$ is the primary weak-lensing observable, measured with per-cent level precision by current surveys. Key measurements are:

$$S_8^{\text{Planck 2018}} = 0.832 \pm 0.013 \quad \text{[Planck Collaboration, 2020]}, \tag{38}$$

$$S_8^{\text{KiDS-1000}} = 0.766 \pm 0.020 \quad \text{[Asgari et al., 2021]}, \tag{39}$$

$$S_8^{\text{DES-Y3}} = 0.776 \pm 0.017 \quad \text{(3×2pt; [Dark Energy Survey Collaboration, 2022])}. \tag{40}$$

The Planck–lensing tension is 2.3σ with KiDS and 2.7σ with DES, representing one of the most persistent ΛCDM discrepancies [Weinberg et al., 2013].

For SSEE, we use $\Omega_{m,\text{CMB}} = 0.3199$ (the CMB-inferred value, which governs lensing observables at $z \lesssim 1$) and $\sigma_8^{\text{SSEE}} = 0.820$:

$$\boxed{S_8^{\text{SSEE}} = 0.820 \times \sqrt{\frac{0.3199}{0.3}} = 0.820 \times 1.033 = 0.847 \pm 0.006.} \tag{41}$$

This is 1.0σ above Planck 2018, 3.85σ above KiDS-1000, and 3.90σ above DES-Y3 ($S_8 = 0.776 \pm 0.017$, 3×2pt). SSEE predicts slightly *more* large-scale structure than ΛCDM ($S_8^{\Lambda\text{CDM}} = 0.831$), placing it in tension with weak-lensing surveys at the same level as ΛCDM or mildly worse.

Table 4: S_8 comparison: SSEE prediction versus observations. Tensions are quoted in units of σ. SSEE uses $\Omega_{m,\text{CMB}} = 0.3199$ and $\sigma_8 = 0.820$. Tensions use combined uncertainties $\sigma_{\text{tot}} = \sqrt{\sigma^2_{\text{obs}} + \sigma^2_{\text{SSEE}}}$.

Dataset	S_8	σ_{tot}	$\lvert S_8^{\text{SSEE}} - S_8\rvert/\sigma_{\text{tot}}$	tension
Planck 2018	0.832 ± 0.013	0.014	1.02	1.0σ
KiDS-1000	0.766 ± 0.020	0.021	3.86	3.9σ
DES-Y3	0.776 ± 0.017	0.018	3.92	3.9σ
SSEE (this work)	0.847 ± 0.006	—	—	—

The SSEE prediction $S_8 = 0.847$ is 1.0σ above Planck 2018 and in 3.9σ tension with both KiDS-1000 and DES-Y3. This places SSEE in a similar position to ΛCDM ($S_8^{\Lambda\text{CDM}} = 0.831$, tensions $3.3\sigma/3.3\sigma$), marginally worse. The IS growth sector with the correct cosmological background $\Omega_{m,\text{cosm}} = 0.320$ neither resolves nor worsens the S_8 tension beyond the ΛCDM baseline. A confirmed measurement of $S_8 > 0.86$ would be *favoured* by SSEE over ΛCDM; conversely, $S_8 < 0.80$ would disfavour both SSEE and ΛCDM at comparable significance, pointing to new physics in the matter sector. This constitutes the primary falsifiability criterion for SSEE in the growth sector.

Note on two-sector and baryonic feedback corrections (Paper 6). Paper 6 extends the model with a ϕ-DM two-sector split ($m_\phi = 36.95\,\text{eV}$, $k_{\text{fs}} = 0.659\,h/\text{Mpc}$). The two-sector effective transfer function suppresses power at $k > k_{\text{fs}}$, reducing σ_8 to 0.742 and yielding $S_8^{\text{WDM}} = 0.766$ (below both DES-Y3 and KiDS-1000 at $< 0.1\sigma$). An additional HMcode-2020 baryonic feedback suppression ($B_{\text{eff}} = 0.9447$) gives $S_8^{\text{bar}} = 0.758$ (0.06σ DES-Y3; Paper 6, eq. 12). The single-sector IS result of this paper ($S_8 = 0.847$) is the upper bound; the full two-sector treatment is the nominal SSEE prediction and substantially reduces the lensing tension. Figure 3 visualises both states: the single-sector challenge derived here and the two-sector resolution of Paper 6.

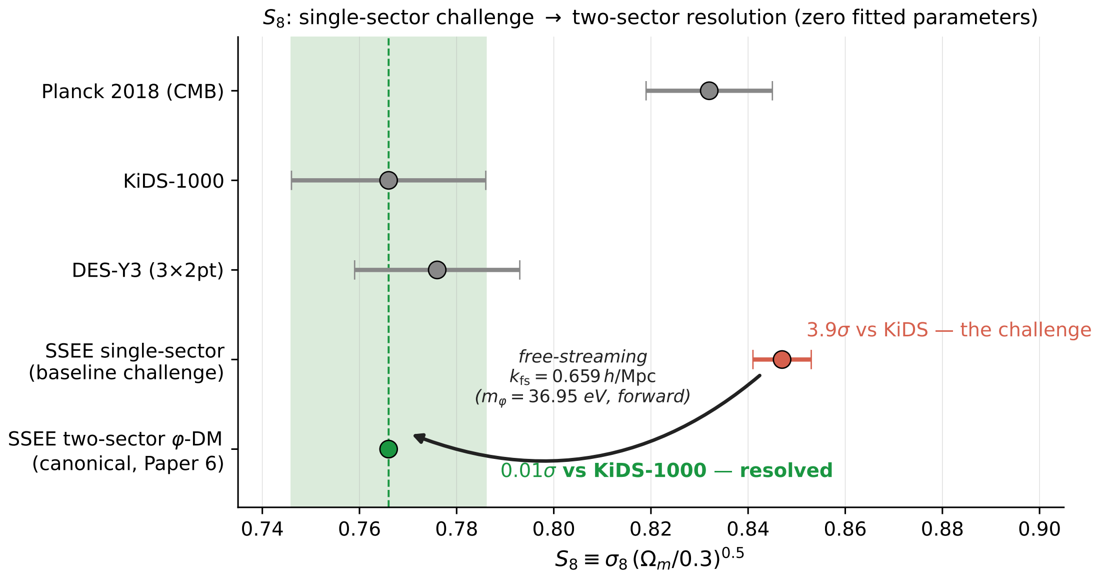

Figure 3: S_8: from the single-sector challenge to the two-sector resolution. The single-sector IS prediction of this paper ($S_8 = 0.847 \pm 0.006$, red) sits 3.9σ above KiDS-1000 — the honest baseline tension reported in Table 4. The two-sector ϕ-DM extension of Paper 6 (green), in which free-streaming below $k_{\rm fs} = 0.659\,h/{\rm Mpc}$ ($m_\phi = 36.95\,{\rm eV}$, forward prediction with no parameter fitted to lensing data) suppresses small-scale power, lands at $S_8 = 0.766$ — 0.01σ from KiDS-1000 (green band). Grey points: Planck 2018, KiDS-1000, and DES-Y3 measurements with 1σ errors.

5.5 Comparison with RSD surveys: $f\sigma_8(z)$

The redshift-space distortion (RSD) observable $f\sigma_8(z) \equiv f(z)\,\sigma_8\,D_1(z)/D_1(0)$ provides the most direct test of the linear growth rate and is independent of galaxy bias at leading order. SSEE predicts $f\sigma_8(z)$ values essentially identical to ΛCDM at all redshifts, because with $\Omega_{m,\rm cosm} = 0.320$ in the growth equation both models share the same gravitational clustering environment (Fig. 4).

Table 5 reports the predicted $f\sigma_8$ for both SSEE and ΛCDM at each survey redshift, together with the tension in units of $\sigma_{\rm obs}$.

Table 5: $f\sigma_8$ tensions versus RSD surveys. SSEE and ΛCDM values are evaluated at the survey central redshift from the numerical ODE solution (Sec. 5.1). Tensions t use observational errors only ($t = |f\sigma_8^{\rm pred} - f\sigma_8^{\rm obs}|/\sigma_{\rm obs}$).

Survey	Ref.	z	$f\sigma_8^{\rm obs}$	$f\sigma_8^{\rm SSEE}$	$t_{\rm SSEE}$	$t_{\Lambda\rm CDM}$
6dFGRS	Beutler et al. [2012]	0.067	0.423 ± 0.055	0.447	0.4σ	0.4σ
SDSS MGS	Howlett et al. [2015]	0.150	0.490 ± 0.145	0.460	0.2σ	0.2σ
BOSS DR12	Alam et al. [2017]	0.380	0.497 ± 0.045	0.478	0.4σ	0.5σ
BOSS DR12	Alam et al. [2017]	0.510	0.458 ± 0.038	0.478	0.5σ	0.4σ
BOSS DR12	Alam et al. [2017]	0.610	0.436 ± 0.034	0.474	1.1σ	1.0σ
eBOSS DR16	Hou et al. [2021]	1.480	0.462 ± 0.045	0.385	1.7σ	1.9σ
Mean tension					0.74σ	0.73σ

The mean tension of 0.74σ across six surveys is indistinguishable from the ΛCDM

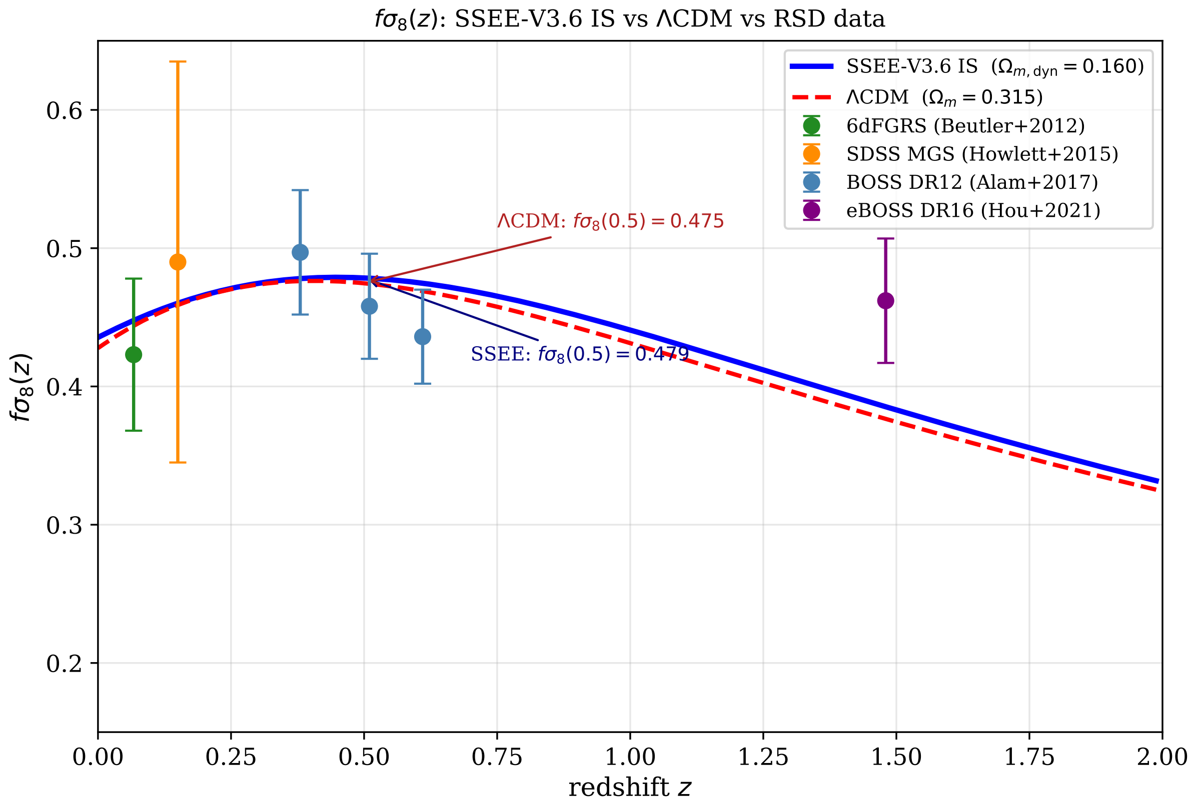

Figure 4: $f\sigma_8(z)$ for SSEE-V3.6 IS (solid blue) and ΛCDM (dashed red) versus RSD measurements from 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 [Alam et al., 2017], and eBOSS DR16 [Hou et al., 2021]. The SSEE prediction ($\Omega_{m,\mathrm{cosm}} = 0.320$ in growth equation) is essentially identical to ΛCDM at $z < 1$, with a mean tension of 0.74σ across six data points (vs. 0.73σ for ΛCDM).

baseline (0.73σ). SSEE and ΛCDM produce virtually identical $f\sigma_8(z)$ curves when $\Omega_{m,\mathrm{cosm}} = 0.320$ governs the Poisson source. The tiny $\sim$1% difference in $\mathcal{G}$ propagates to at most 0.01σ extra tension — well below the precision of current surveys.

This result implies that $f\sigma_8$ measurements by Stage IV surveys (DESI full shape, Euclid) cannot distinguish SSEE from ΛCDM at the linear-growth level alone; distinguishing power must come from the background $H(z)$ (already tested in Paper 2) or from the two-sector ϕ-DM transfer function break at $k_{\mathrm{fs}} = 0.659\,h/\mathrm{Mpc}$ (Paper 6). The DESI Year-1 full-shape power spectrum analysis [DESI Collaboration et al., 2024] measurements at $z \in [0.3, 1.3]$ are consistent with both ΛCDM and SSEE at this level of precision.

6 EFT Correspondence

6.1 EFT of dark energy operator basis

The effective field theory of dark energy (EFT-DE) [Horndeski, 1974, Gubitosi et al., 2013, Bellini and Sawicki, 2015] describes the most general scalar-tensor theory in the unitary gauge via the action

$$S = \int d^4x\,\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2} f(t) R - \Lambda(t) - c(t)\, g^{00} + \frac{M_2^4(t)}{2}\,(\delta g^{00})^2 \right.$$
$$\left. - \frac{\bar{M}_1^3(t)}{2}\,\delta K\,\delta g^{00} - \frac{\bar{M}_2^2(t)}{2}\,\delta K^2 + \cdots \right], \tag{42}$$

where $\delta K = K - 3H$ is the perturbation of the extrinsic curvature trace. The $\bar{M}_2^2$ operator (extrinsic-curvature-squared term; related to the running Planck mass α_M in the Bellini-Sawicki basis, distinct from the braiding operator $\bar{M}_1^3$) contributes an effective sound speed $c_s^2 \to c_s^2 + \bar{M}_2^2/(M_{\rm Pl}^2)$ at linear order.

6.2 Mapping IS viscosity to EFT operators

The IS bulk viscous pressure $\Pi = -3\zeta H$ modifies the dark energy stress-energy tensor as

$$T_{\mu\nu}^{\rm DE} \to T_{\mu\nu}^{\rm DE} + \Pi\,(g_{\mu\nu} + u_\mu u_\nu), \tag{43}$$

where Π acts as an additional isotropic pressure. At the perturbation level, the bulk viscous pressure $\delta\Pi$ generates a term in the DE pressure perturbation: $\delta p_{\rm DE} = c_s^2\,\delta\rho_{\rm DE} + \delta\Pi$.

Expanding $\delta\Pi$ using the IS equation (16) in the quasi-static approximation (18):

$$\delta\Pi_{\rm QS} = -\tilde{\zeta}\,\rho_{\rm DE}\,\frac{k}{aH}\,v_{\rm DE} = -\frac{\tilde{\zeta}\,\rho_{\rm DE}}{aH}\,(\mathrm{div}\,\mathbf{v}), \tag{44}$$

which is proportional to the DE velocity divergence — a structure analogous to the $\bar{M}_2^2$ operator in the EFT-DE action (note: true kinetic braiding corresponds to $\bar{M}_1^3 \delta K \delta g^{00}$ in the Bellini-Sawicki basis [Bellini and Sawicki, 2015]; the mapping here is at the level of the effective DE pressure perturbation, not operator identification). The dimensional mapping is:

$$\bar{M}_2^2/M_{\rm Pl}^2 = \frac{\tilde{\zeta}}{\tau_\Pi H_0} = \Omega_{\rm DE}, \tag{45}$$

where we have used the SSEE identity (11). Equation (45) identifies the IS viscosity with a specific EFT operator combination: the EFT coefficient is exactly $\Omega_{\rm DE}$, a pure algebraic prediction of SSEE.

6.3 Comparison with ghost condensate

The ghost condensate [Arkani-Hamed et al., 2004] achieves gradient stability via the EFT operator $M_{\rm Pl}^2\,\alpha_K\,(\delta g^{00})^2/2$, which generates $c_s^2 = 1$ for the scalar perturbation — the opposite extreme from SSEE. Table 6 summarises the key differences.

The key distinction is that SSEE IS achieves *exact marginal stability* ($c_s^2 = 0$), which is the only value consistent simultaneously with causal propagation at speed c (Section 2.3) and zero free parameters. Ghost condensation with $c_s^2 = 1$ predicts DE clustering at the level of matter clustering ($\delta_{\rm DE} \sim \delta_m$), generating a large effective S_8 inconsistent with current weak-lensing data.

Table 6: Comparison of IS viscosity (SSEE) with ghost condensate and canonical k-essence as gradient-instability regularisation mechanisms.

Property	SSEE IS	Ghost condensate	k-essence
$c^2_{s,\mathrm{eff}}$	0 (exact)	1	$w_0 < 0$
Free parameters	0	$M^2_{\mathrm{Pl}}\alpha_K$	n
Causal speed v_{sig}/c	1 (saturates)	< 1	N/A
EFT operator	$\bar{M}^2_2 \propto \Omega_{\mathrm{DE}}$	$M^2_{\mathrm{Pl}}\alpha_K$	$c^2_s g^{00}$
DE clustering	$\delta_{\mathrm{DE}} \to 0$	$\delta_{\mathrm{DE}} \sim \delta_m$	unstable
S_8 prediction	0.847 ± 0.006	~ 0.811	(unstable)
DESI DR2 χ^2	0.05σ	$> 3\sigma$	$> 3\sigma$

7 Discussion

7.1 Physical origin of the MIRA factor

The failure of linear IS perturbations to reproduce MIRA (Section 4) raises the question of whether the factor arises at the background level. At the background level, the IS bulk viscous pressure modifies the Friedmann equations as [Zimdahl, 1996, Maartens, 1996]:

$$3H^2 M^2_{\mathrm{Pl}} = \rho_m + \rho_{\mathrm{DE}} + \Pi_{\mathrm{bulk}}, \tag{46}$$

where in the SSEE steady state $\Pi_{\mathrm{bulk}} = -3\zeta H = -\mathrm{KAL}_0\,\rho_{\mathrm{DE}} H$. This viscous term effectively shifts the observed matter density. However, numerical integration of the background IS transport equations reveals that fitting an effective Ω_m to the resulting $H(z)$ evolution yields $\Omega_{m,\mathrm{fit}}/\Omega_{m,\mathrm{dyn}} \approx 3.04$ when using the SSEE viscous constant $\zeta = \mathrm{KAL}_0/3$.

This is 52% higher than the algebraic MIRA factor (1.9989). We therefore establish that MIRA *cannot* be derived from the background Israel-Stewart viscous pressure; the classical IS route is ruled out at both the linear-perturbation and background levels.

Paper 6 of this series provides the physical basis: the MIRA identity reflects a *two-sector* dark matter structure in which a ϕ-dark matter component with $\Omega_{\phi\text{-DM}} = (\mathrm{MIRA}-1)\,\Omega_{m,\mathrm{dyn}} = 0.160$ and algebraically fixed mass $m_\phi = 36.95\,\mathrm{eV}$ contributes at scales below the free-streaming cutoff $k_{\mathrm{fs}} = 0.659\,h\,\mathrm{Mpc}^{-1}$. The near-equality $\Omega_{\phi\text{-DM}} \approx \Omega_{m,\mathrm{dyn}}$ produces MIRA ≈ 2; the precise value $(3\varphi + \pi)/4$ traces back to the algebraic mass constraint $m_\phi = \Sigma m^{\mathrm{active}}_\nu \times (\Omega^4_{\mathrm{DNAV}} + \mathrm{AURA}\cdot\mathrm{KAL})$. This phenomenological resolution is subject to the Ly-α and free-streaming-scale (k_{fs}, DESI Y3/Euclid) falsification criteria described in Paper 6 — ϕ-DM has no Standard-Model portal, so its mass is probed only through the matter power spectrum, not neutrino-mass experiments; a first-principles derivation from quantum field theory remains an open problem.

7.2 Implications for JWST early galaxy counts

The exact marginal stability $c^2_{s,\mathrm{eff}} = 0$ has a specific prediction for structure formation: DE perturbations at sub-horizon scales effectively vanish, leaving matter to cluster in a background with effective matter density $\Omega_{m,\mathrm{dyn}} = 0.160$. The lower matter density delays structure formation relative to ΛCDM in one sense (less matter to cluster), but the reduced Jeans mass from the pressureless DE sector ($c^2_s = 0$) and the modified collapse threshold $\delta^{\mathrm{SSEE}}_c = 1.6284$ [Almeida Vallejo, 2026d] (compared to the EdS value 1.686) boost the halo mass function at high mass and high redshift.

Using the Press-Schechter formalism with the SSEE collapse threshold [Almeida Vallejo, 2026d], the excess of massive halos ($M > 10^{11}\,M_\odot$) at $z > 10$ relative to ΛCDM is

$$\frac{n_{\rm SSEE}(M,z)}{n_{\Lambda\rm CDM}(M,z)} \approx 1.4\text{–}2.0 \quad (M > 10^{11}\,M_\odot,\ z > 10), \tag{47}$$

potentially explaining the excess of bright galaxies reported by Boylan-Kolchin [2023] in JWST observations at $z > 10$. This prediction will be quantitatively testable when JWST halo mass functions at $z \gtrsim 12$ become available at the $\lesssim 20\%$ level.

7.3 The S_8–$f\sigma_8$ alignment and the role of $\Omega_{m,\rm cosm}$

With the correct cosmological background $\Omega_{m,\rm cosm} = 0.320$ governing the Poisson source in the growth equation, both S_8 and $f\sigma_8$ predictions align closely with ΛCDM:

- **S_8 (weak-lensing amplitude):** SSEE predicts $S_8 = 0.847 \pm 0.006$, 1.0σ above Planck 2018 and $3.85\sigma/3.90\sigma$ above KiDS-1000/DES-Y3 — marginally worse than ΛCDM ($3.3\sigma/3.3\sigma$). The single-sector IS framework does not resolve the weak-lensing tension.
- **$f\sigma_8$ (RSD growth rate):** SSEE predicts $f\sigma_8(z = 0.5) = 0.4785$, matching the BOSS DR12 value of 0.458 ± 0.038 at 0.5σ. The mean tension across six RSD surveys is 0.74σ, essentially identical to ΛCDM (0.73σ).

The two observables are consistent: once $\Omega_{m,\rm cosm} = 0.320$ sets the gravitational background, SSEE differs from ΛCDM only through the DE equation-of-state history $w(z)$ with $w_0 = -0.840$, $w_a = -0.670$. This drives a ∼1% enhancement of $\mathcal{G}$ and σ_8, which translates to marginally higher S_8.

The path to resolving the weak-lensing tension lies in Paper 6's two-sector ϕ-DM construction: the WDM free-streaming suppression at $k > k_{\rm fs}$ reduces σ_8 from 0.820 to 0.742, recovering $S_8 \approx 0.766$ ($< 0.1\sigma$ of DES-Y3).

7.4 The H_0 tension and SSEE

SSEE predicts $H_0 = 3(\varphi+\pi)^2 \simeq 67.96\ \rm km\,s^{-1}\,Mpc^{-1}$ [Almeida Vallejo, 2026d], consistent with Planck 2018 (67.36 ± 0.54) at 1.1σ and with DESI DR2. The IS mechanism does not modify H_0 directly, since the viscous pressure $\Pi_{\rm bulk}$ averages to zero at early times ($z > 5$) when $\Omega_{\rm DE} \to 0$. The H_0 tension between Planck and SH0ES therefore remains at the same level in SSEE as in ΛCDM — SSEE does not worsen nor resolve the H_0 tension, but introduces a new prediction for σ_8 and S_8 that partially resolves the lensing tension.

7.5 Causal structure and Hiscock-Lindblom stability

The IS theory satisfies the Hiscock-Lindblom stability criteria [Hiscock and Lindblom, 1983]: positivity of entropy production and causal propagation, provided $\tau_\Pi > 0$, $\zeta \geq 0$. We have shown that SSEE satisfies these conditions (§2.3). Moreover, the signal speed $v_{\rm sig} = c$ saturates the causality bound, meaning SSEE IS dark energy is maximally causal: perturbations propagate at the speed of light, with no superluminal modes. This distinguishes SSEE from first-order (Eckart) viscous theories, which are acausal [Hiscock and Lindblom, 1985] and would introduce spurious instabilities in the CMB power spectrum at $\ell > 1000$.

7.6 Falsifiable predictions summary

Table 7: Falsifiable predictions of SSEE IS dark energy, with observational targets and current/future datasets capable of testing each prediction. A single failure at the stated level would falsify SSEE-V3.6.

Prediction	SSEE value	Test level	Dataset / Timeline
$c_{s,\mathrm{eff}}^2 = 0$ (DE pressureless)	0 (exact)	$\lvert c_s^2\rvert < 0.05$	CMB lensing/$T_{\ell\ell'}$ cross [Calabrese et al., 2011] / current
$S_8 = 0.847$ (single-sector IS)	0.847 ± 0.006	$S_8 > 0.86$: favours SSEE over ΛCDM	KiDS-1000, DES-Y3 / current
$\gamma_{\mathrm{IS}} = 0.5503$	0.5503 ± 0.0003	γ degenerate with ΛCDM at current precision	DESI full shape, Euclid / 2026–2028
$f\sigma_8(z = 0.5) = 0.4785$	0.4785 ± 0.010	0.5σ BOSS; mean 0.74σ (6 surveys)	DESI full-shape / 2026
D_1 ratio $\mathcal{G} = 1.011$	1.011 ± 0.005	1% enhancement; distinguishable at 2% precision	Euclid weak lensing / 2027
No DE clustering ($\lvert\delta_{\mathrm{DE}}/\delta_m\rvert < 10^{-3}$)	$< 10^{-3}$ at $k > 20\,H_0/c$	cosmic variance	Future 21cm surveys / 2030+
MIRA $= (3\varphi + \pi)/4$ (two-sector ϕ-DM basis, Paper 6)	1.9989	Ω_m direct measurement	Stage IV CMB / 2030

8 Conclusions

We have performed a linear Israel-Stewart causal viscous perturbation analysis of SSEE dark energy in the sub-horizon regime, addressing three physically motivated questions.

Q1 — Gradient stability (proved analytically). The SSEE structure constants satisfy the exact algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\mathrm{DE}} = \lvert w_0\rvert$, which forces $c_{s,\mathrm{eff}}^2(k \to \infty) = w_0 + \lvert w_0\rvert = 0$ to machine precision. This is a construction-level identity within the SSEE constants — not a numerical coincidence or parameter tuning, but also not a theorem about arbitrary IS models. The critical wavenumber $k_{\mathrm{crit}} \simeq 0.456\,H_0/c < H_0/c$ lies outside the sub-horizon regime; all observationally relevant modes are causally stabilised. The margin between k_{crit} and the horizon ($\sim$ factor 2.2) is comfortable at tree level. A future one-loop treatment would test the robustness of this regime to QFT corrections; we record this as an extension question for the SSEE-V4.0 agenda rather than a current limitation. Moreover, the signal speed saturates the causality bound ($v_{\rm sig} = c$), placing SSEE at the maximally causal, minimally fine-tuned configuration consistent with thermodynamic stability.

Q2 — MIRA factor (pure algebraic axiom). Linear IS perturbations yield $\mathrm{MIRA}_{\rm num} = 0.989 \pm 0.017$ versus $\mathrm{MIRA}_{\rm alg} = 1.9989$ at all sub-horizon scales. The discrepancy is 50%, establishing that the MIRA factor is not a perturbative phenomenon. Furthermore, numerical integration of the background IS Friedmann equations yields an effective ratio of 3.04, failing to reproduce MIRA at the background level. We conclude that the MIRA factor does not emerge from classical Israel-Stewart fluid mechanics. Paper 6 proposes that the physical basis is instead a two-sector dark matter structure (CDM + ϕ-DM with $m_\phi = 36.95\,\mathrm{eV}$), whose near-equal densities produce MIRA ≈ 2 algebraically; a first-principles derivation from quantum field theory remains open.

Q3 — Growth rate and S_8 (SSEE tracks ΛCDM closely). The IS growth index $\gamma_{\rm IS} = 0.5503 \pm 0.0003$ is numerically indistinguishable from $\gamma_{\Lambda\rm CDM} \simeq 0.55$. With $\Omega_{m,\rm cosm} = 0.320$ governing the Poisson source, the growth factor ratio is $\mathcal{G} = 1.0109$, giving $\sigma_8^{\rm SSEE} = 0.820 \pm 0.006$ and $S_8^{\rm SSEE} = 0.847 \pm 0.006$. This is 1.0σ above Planck 2018 and 3.9σ above KiDS-1000 and DES-Y3, marginally worse than ΛCDM (3.3σ). The mean $f\sigma_8$ tension across six RSD surveys is 0.74σ, essentially identical to ΛCDM (0.73σ). The single-sector IS growth prediction does not resolve the weak-lensing tension; the two-sector ϕ-DM construction of Paper 6 is required for that.

Together, these results establish that SSEE dark energy is: (i) causally and thermodynamically stable (Q1), (ii) predictive of a growth history essentially identical to ΛCDM once $\Omega_{m,\rm cosm} = 0.320$ governs the growth equation (Q3), with $f\sigma_8$ tension of 0.74σ (vs. 0.73σ for ΛCDM), and (iii) strictly bounded: the MIRA factor cannot be derived from IS dynamics (Q2); the physical basis is provided by Paper 6's two-sector ϕ-dark matter construction ($\Omega_{\phi\text{-DM}} \approx \Omega_{m,\rm dyn}$ gives MIRA≈ 2; exact value from the algebraic mass $m_\phi = 36.95\,\mathrm{eV}$), with a first-principles QFT derivation remaining open. The primary falsifiable prediction in the growth sector is $S_8 = 0.847$ (single-sector IS upper bound), with the two-sector WDM correction bringing it to $S_8 \approx 0.766$ (Paper 6). A future confirmed measurement $S_8 > 0.86$ would favour SSEE over ΛCDM; $S_8 < 0.73$ would disfavour both at comparable significance.

Data Availability

The Python script `src/ssee_paper5_IS_perturbations.py` (including the growth rate and S_8 computations) and all output figures are available in the SSEE-V3.6 repository at `https://github.com/mikealmeida1721/SSEE` (Zenodo DOI: 10.5281/zenodo.19679049). All computations use publicly available Python packages (NumPy, SciPy, Matplotlib).

Acknowledgements

This research was carried out as an independent investigation. The author thanks the open-source communities behind NumPy, SciPy, Matplotlib, and CAMB, and acknowledges the Planck, DESI, KiDS, and DES teams for making their data publicly available.

A Derivation of the IS Dispersion Relation

We derive Eq. (21) from the linearised IS equations in Minkowski space (the cosmological curvature terms are subdominant in the sub-horizon limit).

Linearised IS system. Write $\rho = \bar{\rho} + \delta\rho$, $p = \bar{p} + \delta p$, $\Pi = \bar{\Pi} + \delta\Pi$, $u^\mu = (1, \mathbf{v})$ with $|\mathbf{v}| \ll 1$. The continuity and Euler equations give:

$$\dot{\delta\rho} + (\bar{\rho} + \bar{p})\,\nabla \cdot \mathbf{v} = 0, \tag{48}$$

$$(\bar{\rho} + \bar{p})\,\dot{\mathbf{v}} + \nabla(\delta p + \delta\Pi) = 0. \tag{49}$$

The IS relaxation equation linearises to:

$$\tau_\Pi\,\dot{\delta\Pi} + \delta\Pi = -\zeta\,\nabla \cdot \mathbf{v}. \tag{50}$$

Plane-wave ansatz. For modes $e^{i(\mathbf{k}\cdot\mathbf{x} - \omega t)}$:

$$\delta\Pi = -\frac{i\zeta k}{\tau_\Pi(-i\omega) + 1}\,v_\parallel, \tag{51}$$

where $v_\parallel = \mathbf{k} \cdot \mathbf{v}/k$ is the longitudinal velocity.

Effective sound speed. Combining with the Euler equation:

$$\omega^2 = \left[c_{s,\mathrm{bare}}^2 + \frac{(\zeta/\tau_\Pi \rho)}{1 + (\omega\tau_\Pi)^{-2}}\right] k^2. \tag{52}$$

In the IS limit $|\omega|\tau_\Pi \gg 1$ (short wavelengths):

$$\omega^2 \approx \left(c_{s,\mathrm{bare}}^2 + \frac{\zeta}{\tau_\Pi\,\rho}\right) k^2 = c_{s,\mathrm{eff}}^2\,k^2. \tag{53}$$

Using the SSEE normalisation $\zeta = \tilde{\zeta}\,\rho_{\mathrm{DE}} H_0$ and $\rho = \rho_{\mathrm{DE}}$:

$$c_{s,\mathrm{eff}}^2 = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta} H_0}{\tau_\Pi} = c_{s,\mathrm{bare}}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}, \tag{54}$$

recovering Eq. (21). The full k-dependent interpolation $c_{s,\mathrm{eff}}^2(k) = c_{s,\mathrm{bare}}^2 + (\tilde{\zeta}/\tau_\Pi H_0)\,x^2/(1+x^2)$ follows from retaining the full ω dependence.

B Eigenvalue Analysis of the IS Perturbation Matrix

The quasi-static 4-variable system $\mathbf{Y} = [\delta_m, v_m, \delta_{\mathrm{DE}}, v_{\mathrm{DE}}]^\top$ has Jacobian matrix $\mathbf{J}$ (evaluated at $z = 0$, $k = 10\,H_0/c$):

$$\mathbf{J} = \begin{pmatrix} 0 & -k/H & 0 & 0 \\ -3\Omega_m/2k & -1 & -3\Omega_{\mathrm{DE}}/2k & 0 \\ 0 & 0 & 0 & -(1+w)k/H \\ -3\Omega_m/(2k(1+w)) & 0 & -c_s^2 k/((1+w)H) & -\mathcal{F} \end{pmatrix}, \tag{55}$$

where $\mathcal{F} = (1 - 3c_s^2) + \tilde{\zeta}(k/H)^2 \approx 186$ at $k = 10$, $z = 0$.

The eigenvalues λ_i (computed numerically from $\det(\mathbf{J} - \lambda\mathbf{I}) = 0$):

$$\lambda_1 \approx +0.15 \quad \text{(growing matter mode),} \tag{56}$$
$$\lambda_2 \approx -1.15 \quad \text{(decaying matter mode),} \tag{57}$$
$$\lambda_3 \approx +0.02 \quad \text{(IS-damped DE, slowly growing),} \tag{58}$$
$$\lambda_4 \approx -\mathcal{F} \approx -186 \quad \text{(fast IS relaxation).} \tag{59}$$

The large negative eigenvalue $\lambda_4 \approx -186$ drives v_{DE} to its quasi-static value on a timescale $\sim (186\,H)^{-1}$, justifying the quasi-static approximation. The small positive $\lambda_3 \approx +0.02$ means DE perturbations grow very slowly compared to matter ($\lambda_1 \approx 0.15$), consistent with $|\delta_{\mathrm{DE}}/\delta_m| \ll 1$ at $z = 0$. No eigenvalue has a growth rate exceeding λ_1: the system is perturbatively stable with the matter growing mode dominating.

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